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Restaurant Ordering System

System Overview

Table Ordering: Customers browse menus, and customize orders, right from the app, choosing either take out, delivery, or reserve a table.

Kitchen Integration: Orders stream instantly to kitchen staff, on what orders they should make.

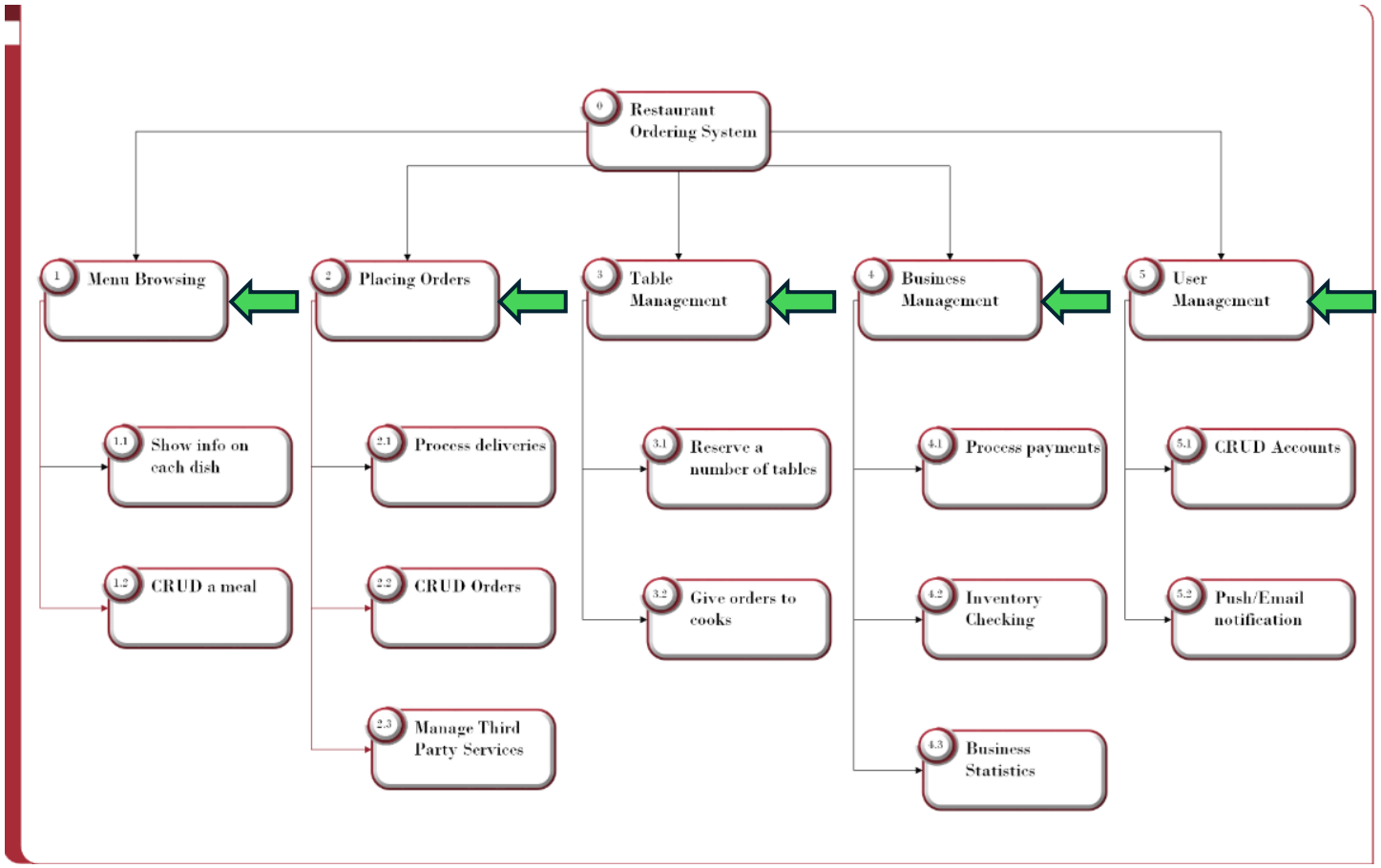
Table & Staff Management: Track table status (occupied/ready) and optimize seating base on time and occupancy.

Business metrics: Track inventory, monitor sales, and generate reports for business insights

Requirement Modeling

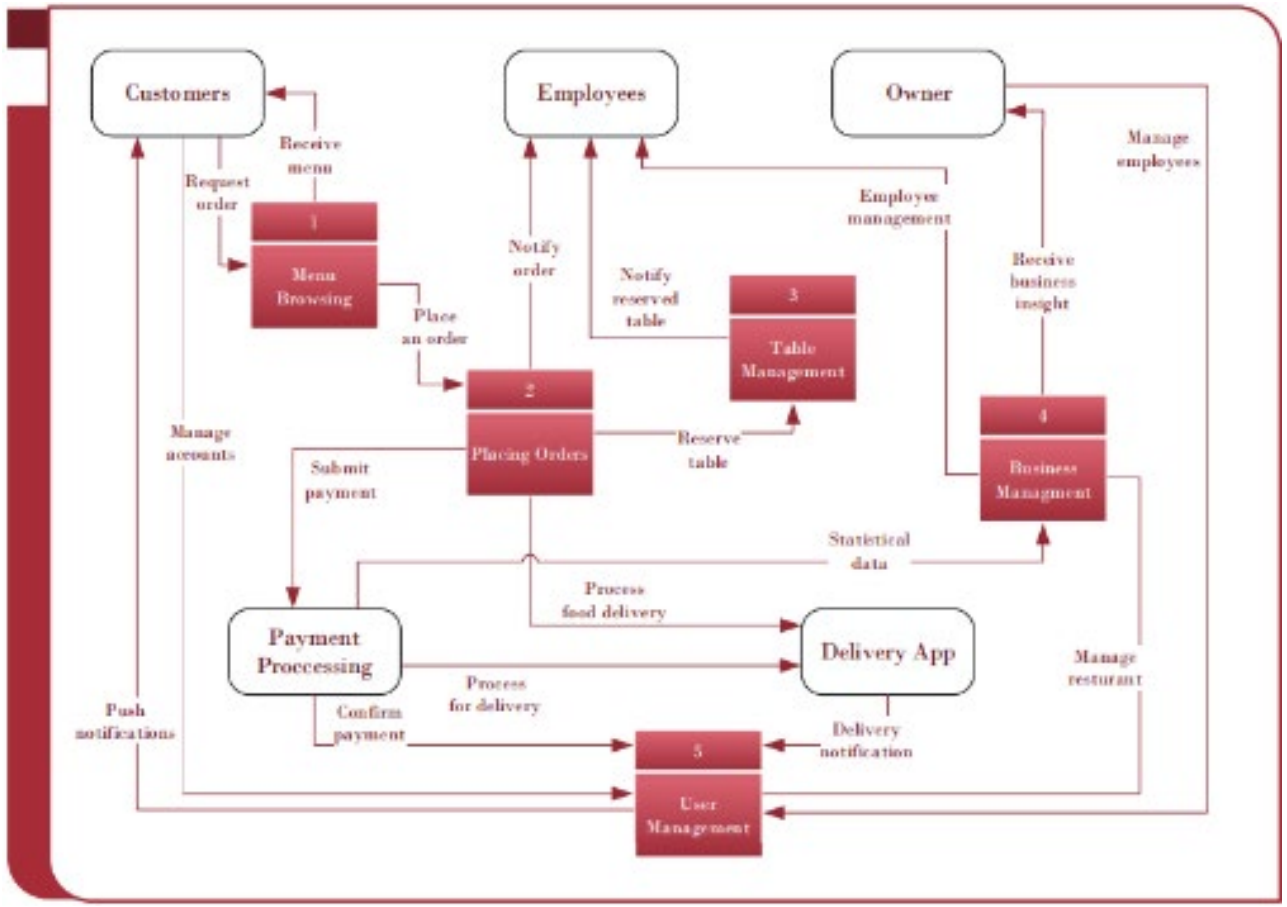
- Customer - menu of the restaurant, - choose meal - option for order takeout or delivery or reserve a table - payment.
- Employees - notified of incoming orders/reservations, receive scheduling changes
- Owner – receives business analytics - ability to change the menu/prices
- Application – manage logic for 3rd party entities like delivery and payment - provide push/email notifications for order status

Functional Decomposition Diagram (FDD)



1. Menu Browsing
   - 1.1 - Show info on each dish
   - 1.2 - CRUD a meal
2. Placing Orders
   - 2.1 - Process deliveries
   - 2.2 - CRUD Orders
   - 2.3 - Manage Third Party Services
3. Table Management
   - 3.1 - Reserve a number of tables
   - 3.2 - Give orders to cooks
4. Business Management
   - 4.1 - Process payments
   - 4.2 - Inventory Checking
   - 4.3 - Business Statistics
5. User Management
   - 5.1 - CRUD Accounts
   - 5.2 - Push/Email notification

Diagram 0



•Process 1 Menu Browsing: Query information about what the restaurant has. Includes info about availability, prices, nutrition, and estimated time to get ready.

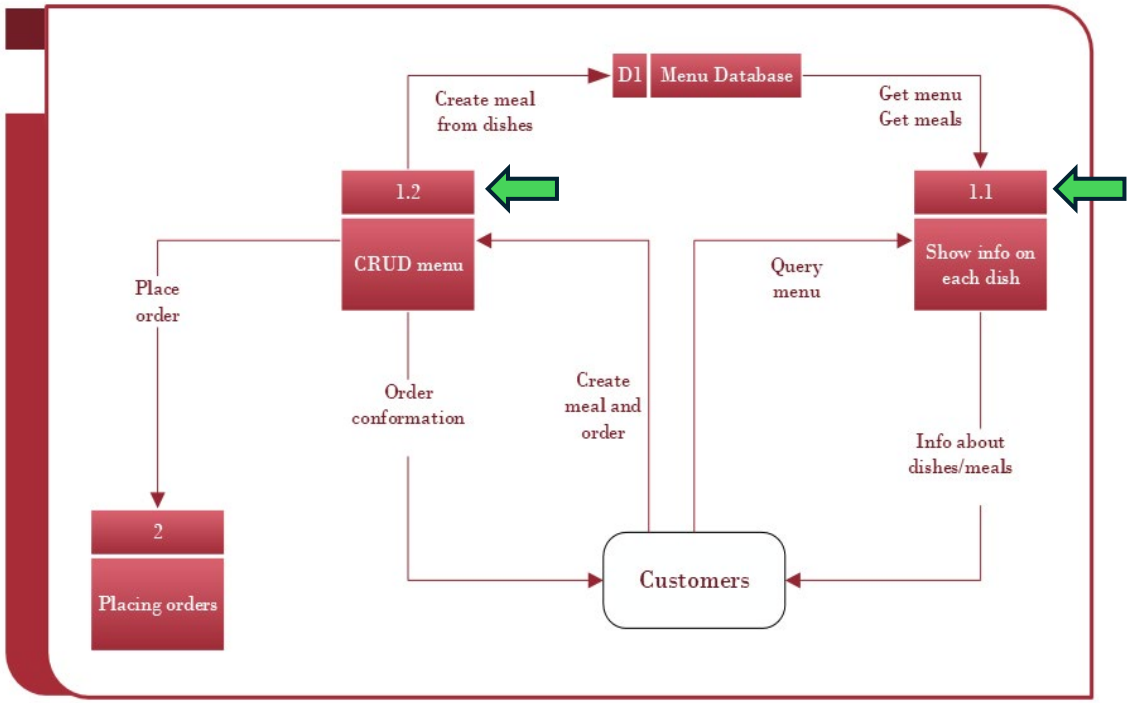
•Process 2 Placing Orders: From the pick-out menu, an order will be queued for the chefs to make and then queued for delivery.

•Process 3 Table Management: Allows users to reserve a pre-paid table soon, accounts for the number of people and places to order.

•Process 4 Business Management: gathers statistics on the

•Process 5 User Management: Allows CRUD accounts, set up email/push notifications, set up payment and have access to the restaurant's services

Diagram 1: Menu Browsing



<u>1.1 Show Info On Each Dish</u>

RECEIVE Menu Query from Customer

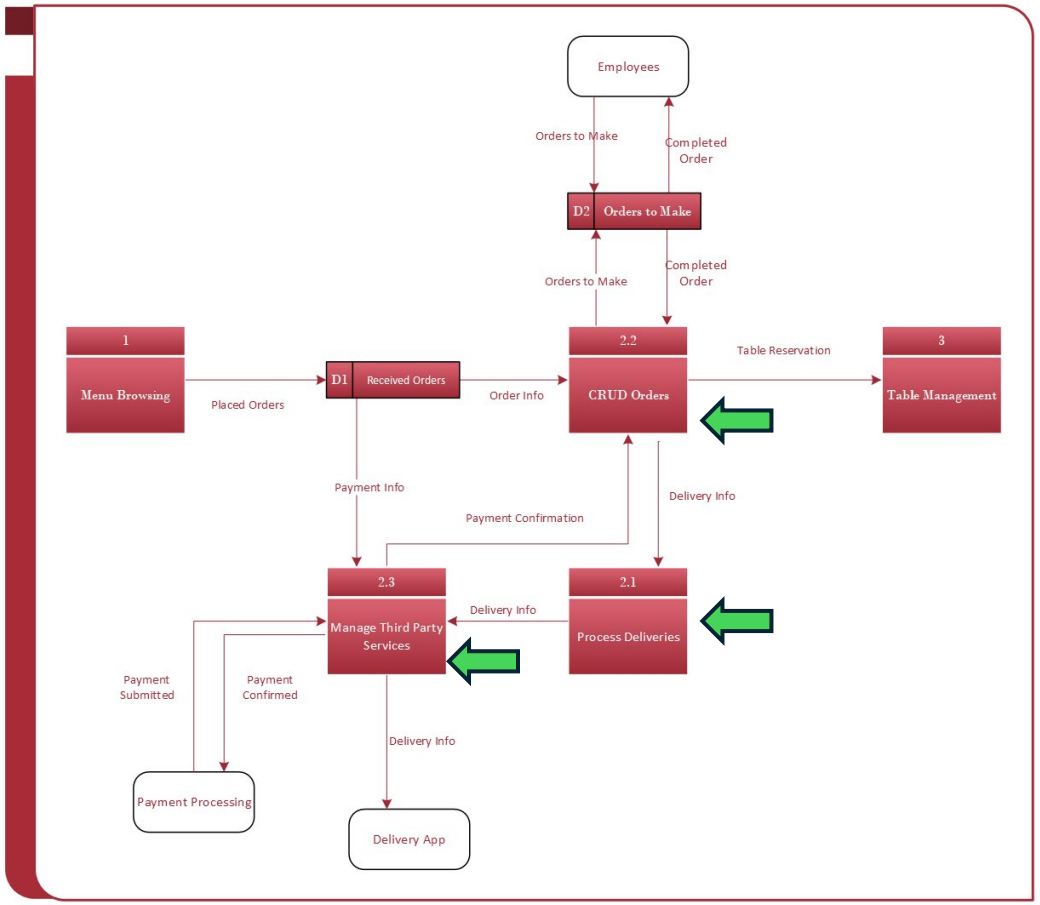
IF Query = “Full Menu” THEN

SEND List of Meals (name, description, price, ingredients, prep time) to Customer

ELSE IF Query = “Specific Item” THEN

SEND Detailed Meal Info (nutritional info, full description) to Customer

Diagram 2: Placing Orders



2.1 Process Deliveries

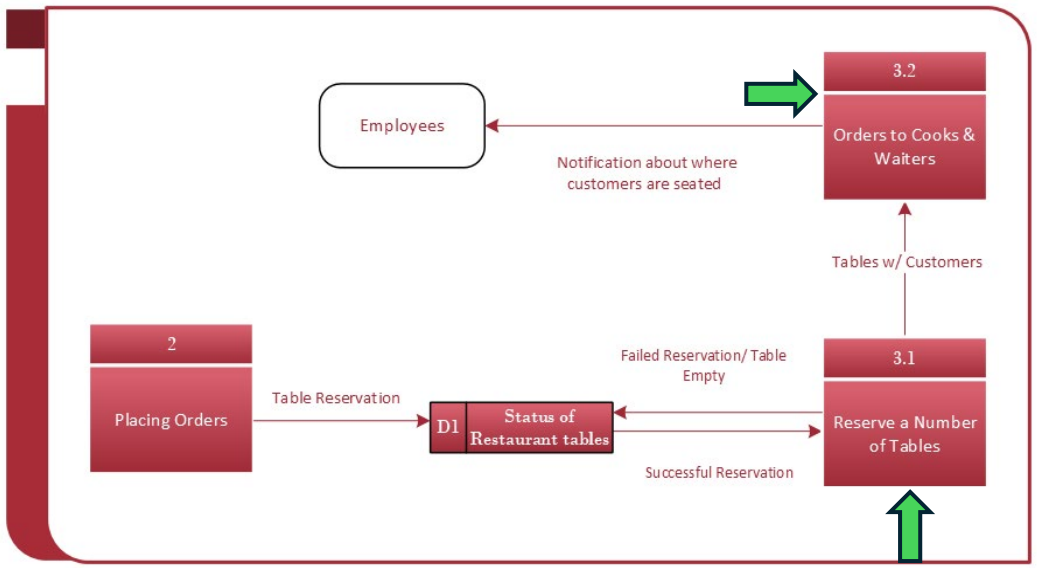
RECEIVE Delivery Info from CRUD Orders (2.2)

IF Order Contains Delivery Details THEN

FORMAT Delivery Info for Delivery App

SEND Delivery Info to Manage Third Party Services (2.3)

Diagram 3: Table Management



3.1 Reserve a Number of Tables

RECEIVE Reservation Request from Table Management (30

RECEIVE Table Status from Status of Restaurant Tables (D1)

IF Table Available THEN

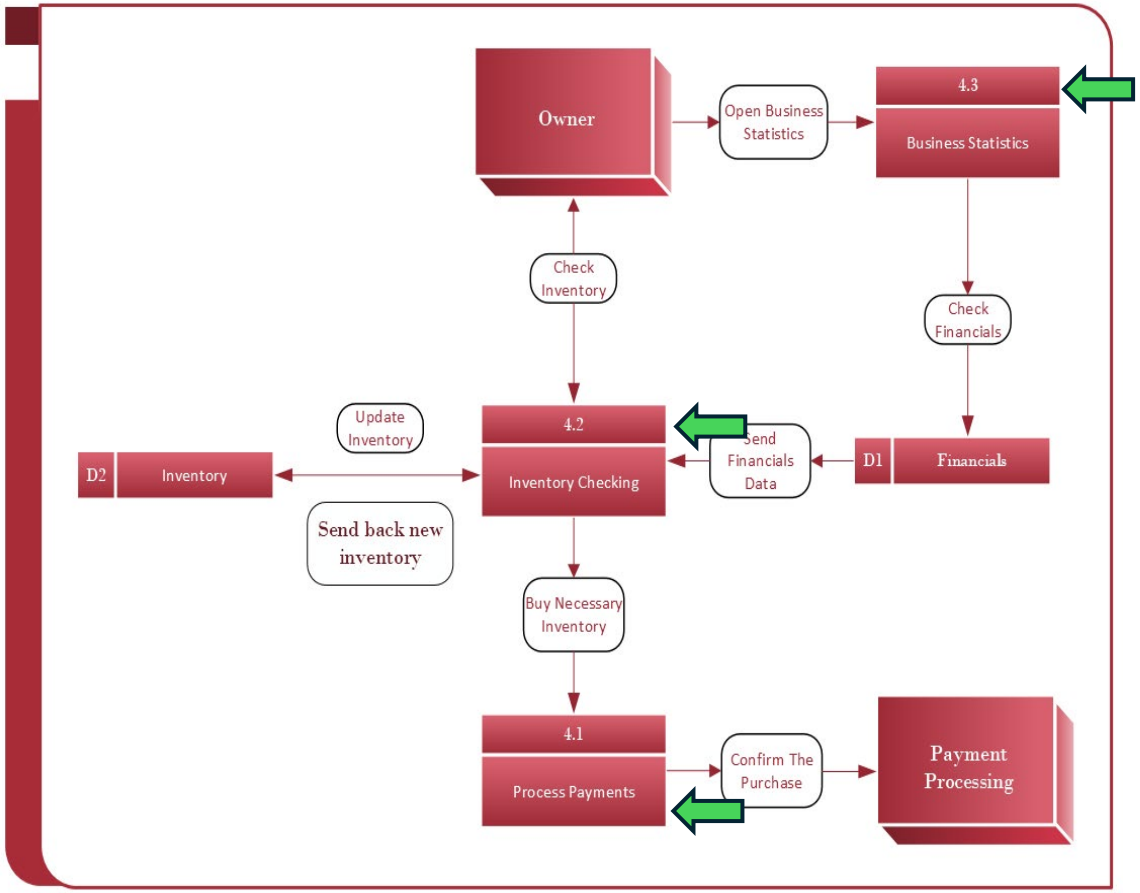
STORE Reservation in Table Database

SEND Table with Customer Info to Give Orders to Cooks (3.2)

ELSE

UPDATE Table Status to “Unavailable” in D1

Diagram 4: Process Payments



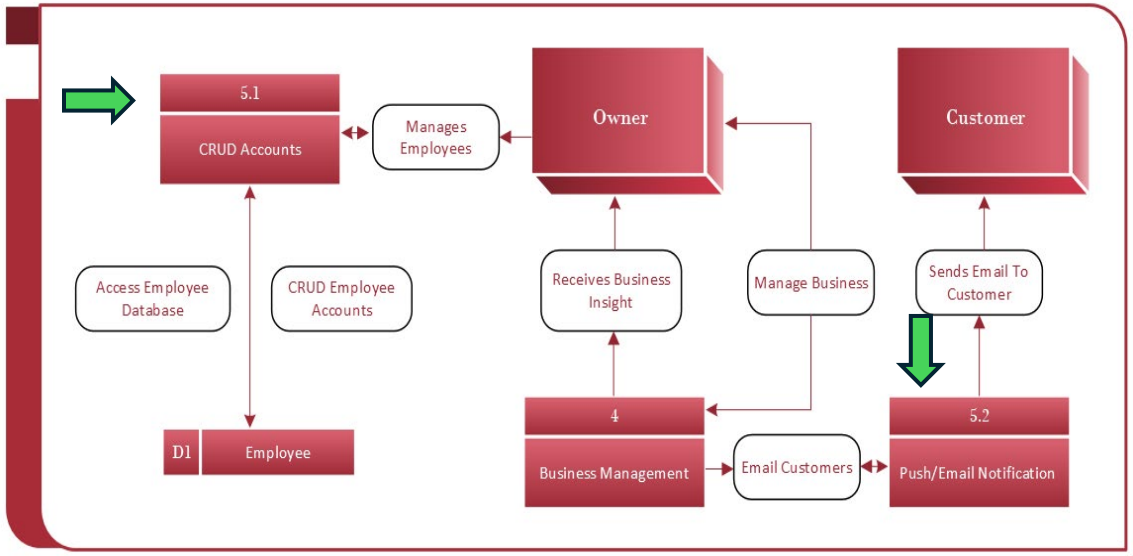
4.1 Process Payments

RECEIVE Current Inventory and Budget from Inventory Database

IF Inventory or Budget Updated THEN

SEND Purchase Request to Inventory Checking (4.2)

Diagram 5: User Management



5.1 CRUD Accounts

RECEIVE Account Update Request from Owner

GET Employee Info from Employee Database

IF Account Exists THEN

UPDATE Employee Record

ELSE

STORE New Employee Record

SEND Account Update Confirmation to Owner

Conclusion

- This restaurant ordering system shows a well-organized and improved efficient dining environment.
- This system is developed with processes like menu browsing, order placement, table management, and inventory reviews on a digital platform.
- Because of this enhanced dining system, errors will be minimized, wait times will be lowered, and the overall restaurant experience for staff, management, as well as customers.